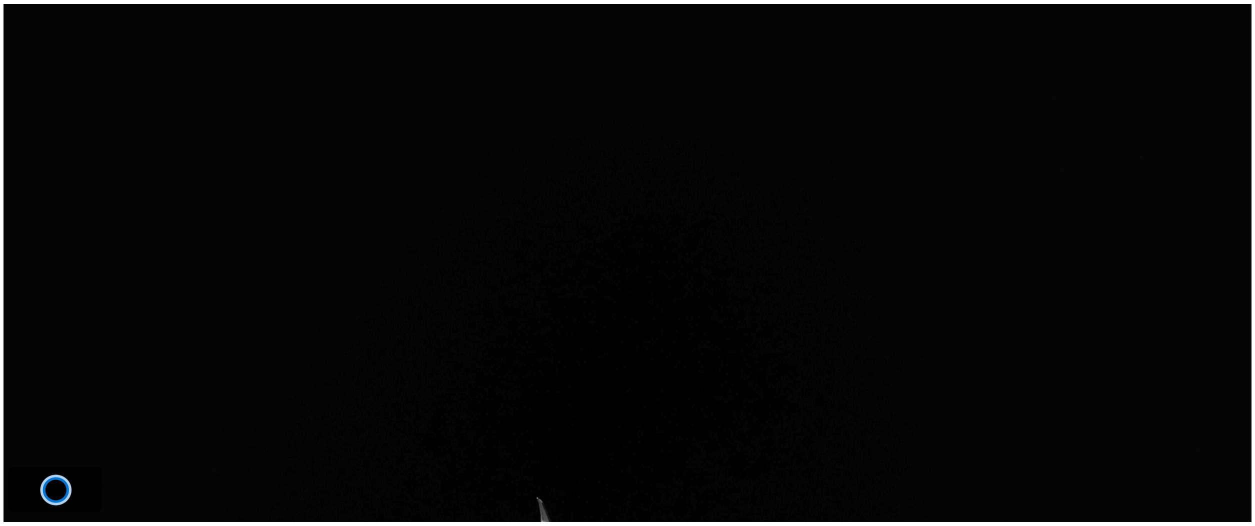

Project charter AISSA

Table of contents

Goal

Define the high-level purpose of the system. Focus on the value it adds to the environment.

- Automatize the reception of people in the office environment and help the employee to schedule meetings with customers/co-workers or receive messages during it's absence.

Description

Provide a brief description of the problem that needs solving and explain how the solution works.

- Develop an embedded IA assistant for office entrance that automates reception tasks, manage messages and pass information from visitors to the office owner. This system, uses an intuitive interface that allows the interaction with the assistant through voice commands. All messages received will be notified to the office owner via instant messages.

Technology Stack

Define which technologies i.e. software and hardware, are to be utilized on the project

- Hardware
  - Embedded system (Raspberry Pi 5)
  - HD camera (Waiting for availability confirmation)
  - Microphone and speaker (Waiting for availability confirmation)
  - Touch or standard display (Waiting for availability confirmation)
- Software
  - Embedded Linux OS
  - Modular, service-oriented architecture
    - Perception Service
      - Manage hardware of detection (Camera, Microphone)
    - Control Service
      - Logic for LLM
    - Frontend
      - Digital display and audio through speaker
    - Backend
      - Database management
- AI and interaction
  - Face detection: OpenCV
  - Speech recognition: Whisper / Google Speech API
  - Text-to-speech: Coqui TTS / Amazon Polly
  - Conversational logic: LLM-based assistant (local or cloud)
- Frontend
  - Digital face rendering: Unity / Three.js
  - UI control via Python or C++ bindings
- Backend
  - Language: Python
  - Local database: SQLite / PostgreSQL / Firebase
  - Messaging/notification: Email or mobile app integration (Telegram)

Success factors

Establish the Project Success Criteria and Acceptance Standards, defining quantifiable metrics and specific conditions (SMART notation) that must be fulfilled to validate satisfactory delivery to stakeholders.

- The system responds within acceptable response times.
- The system is constantly looking for human presence during office hours.
- The system AI assists and interacts with visitors fluently.
- Messages are recorded and sent via text to the office owner.
- The system functionality is user friendly.
- All information received remains stored and encrypted.

Risks

Identify and document the High-Level Project Risks, including potential threats and opportunities, to establish a proactive mitigation strategy and ensure project resilience.

- Threats:
  - Privacy and data protection
    - Since the system captures camera feeds, voice recordings, and visitor messages, there is a risk of violating privacy regulations. We can implement encryption to all the stored information
  - Environmental Factors
    - Variations in lighting or high ambient noise at the office entrance could interfere with the camera's presence detection or the microphone's ability to capture clear voice messages, so we should implement some alerts so the AI knows what's happening.
- Opportunities
  - Scalability of AI Interaction
    - There are new models of LLM
  - Modernization in the office
    - With the implementation of a IA based system, the office will look innovative
  - Improved availability
    - With an AI assistant, availability is secured.

Milestones & schedule (Development plan)

Establish the Project Schedule Baseline, identifying key milestones, current status, and designated owners to ensure alignment with critical deadlines and project constraints.

Timeline

Cycle 1: Core AISSA	Date	Lead	Status
Project initiation and scope definition	@January 30, 2026	MIGUEL ANGEL HERNANDEZ CAMACHO	Completed
Requirements and WBS	@January 30, 2026 → February 1, 2026	MIGUEL ANGEL HERNANDEZ CAMACHO	In progress
Presence detection and greeting	@February 2, 2026 → February 9, 2026		Not started
Digital face rendering	@February 2, 2026 → February 6, 2026	J JESUS PEREZ RODRIGUEZ	Not started
Basic speech input/output	@February 5, 2026 → February 10, 2026		Not started
Testing	@February 9, 2026 → February 13, 2026		Not started
Message storage and notification	@February 16, 2026 → February 20, 2026		Not started

Cycle 1: Core AISSA	Date	Lead	Status
AI conversational refinement	@February 16, 2026 → March 5, 2026	J JESUS PEREZ RODRIGUEZ	Not started
Security and privacy controls	@February 19, 2026 → March 2, 2026		
System integration testing	@February 24, 2026 → March 2, 2026		Not started
User acceptance testing	@February 26, 2026 → March 5, 2026		Not started

Assumptions & constraints

List the known limitations and conditions that are taken for granted.

Known Conditions

- There is total compatibility between needed components.
- Open-source software is sufficient to fulfill the project requirements.
- The environmental conditions are good enough for the system to work without the system being harmed.
- The is a continuous supply of energy and internet.

Known Limitations

- The model rendering can be a tough task for systems with less than 8GB of RAM.
- The Raspberry Pi RAM cannot be upgraded nor extended.
- A total of 30 work hours per week.

Project manager & team members

Define the Project Organizational Structure, detailing the assigned human resources, their specific functional roles, and the matrix of responsibilities required for project execution.

Team

Team member	Role	Responsibility
Miguel Angel Hernández	Leader	Product Manager
Rodrigo Samuel Bernal	Co-leader	Quality Assurance
Enrique Alfonso Gracián	Developer	Developer / Process Engineer
Jesús Pérez	Developer	UI / UX designer
Gustavo Enrique Martínez	Developer	Product Owner
Lucio Emiliano Ruiz	Developer	Developer

Approval

This section is where stakeholders and authorizers can sign off, letting the team know we can get started.

Stakeholders

Stakeholder	Date signed	Approval
JORGE RAFAEL AGUILAR CISNEROS	@February 1, 2026	Not approved